# Range of $\alpha-$particles and setting up $\alpha-\gamma$ coincidence

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The range of alpha particles in air was measured by recording their residual energy at different pressures and obtained an extrapolated range of $(2.61 \pm 0.01)$ cm at STP. We also investigated alpha–gamma coincidences in the decay of $^{241}$Am using pulse-height calibration, baseline RMS cut-off filtering, energy-correlation heatmap analysis, and time-walk correction. The trigger-time and constant-fraction methods yielded half-lives of $(63 \pm 2)$ ns and $(82 \pm 3)$ ns, respectively. The trigger-time estimate is closer to the literature value 67.2 ns of half-life of $^{237}$Np excited state.

## I. INTRODUCTION

Alpha decay and gamma-ray emission provide information about nuclear structure and radioactive decay. In alpha decay, the parent nucleus emits an alpha particle, and the daughter nucleus may be left in an excited state. The range of alpha particles is studied by measuring their residual energy after passing through air at different pressures using a $^{241}$Am source and a silicon surface-barrier detector.

The second part investigates alpha–gamma coincidence in the decay of $^{241}$Am to $^{237}$Np. Simultaneous detection of alpha particles and gamma photons allows correlated events to be identified using their energies and time differences. These measurements help investigate the 59.5 keV transition and determine the half-life of the excited nuclear state.

## II. THEORY

### A. Range of alpha particles

Alpha particles are positively charged helium nuclei consisting of two protons and two neutrons. As they pass through matter, they lose kinetic energy primarily through the excitation and ionisation of atoms and molecules. Due to their relatively large mass, their trajectories remain approximately straight during this process. The total distance travelled before an alpha particle loses all its kinetic energy is called its *range*. It depends on the initial kinetic energy, the density of the absorbing medium, and the atomic number and mean ionisation potential of its atoms [1].

The energy loss is not uniform along the trajectory. As the alpha particle slows down, its interaction time with the electrons of the medium increases. Consequently, the energy loss per unit distance increases towards the end of its path, reaches a maximum, and finally falls to zero when the particle stops. The corresponding variation of specific ionisation with distance is known as the *Bragg curve*.

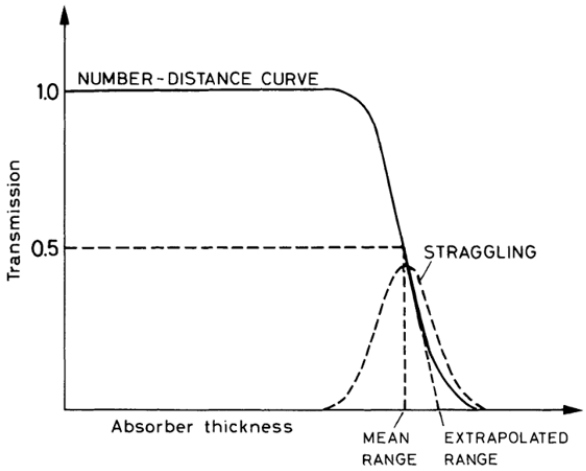



FIG. 1. Illustration of the extrapolation method used to estimate the range of alpha particles from the residual-energy curve. (Adapted from [2])

The range of alpha particles can be determined by measuring their residual energy at the detector as a function of the effective distance travelled through the medium. The range is not perfectly definite, even for alpha particles with identical initial energies, because individual particles undergo different interactions and experience fluctuations in energy loss. This phenomenon is called *range straggling*. Consequently, the residual-energy curve exhibits a tail near the endpoint, and the range is estimated by extrapolating the appropriate curve to its endpoint [2].

When the geometrical separation between the source and detector is fixed, the effective distance travelled through air can be varied by changing the chamber pressure. The effective distance is defined as the distance in standard air containing the same number of air molecules as the actual air column between the source and detector. It is given by

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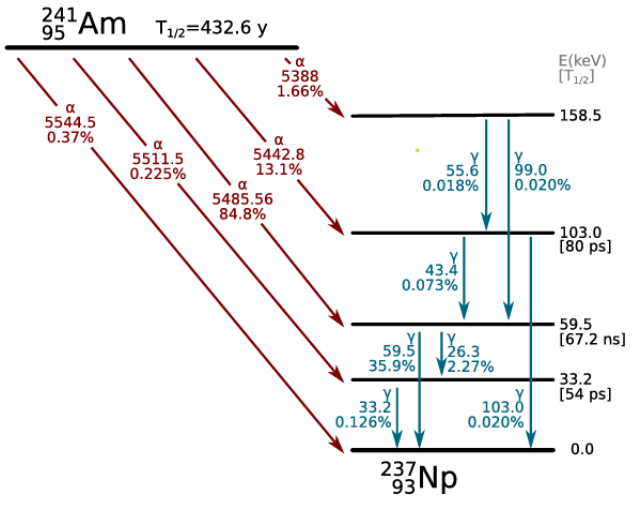



FIG. 2. Decay scheme of $^{241}$Am showing alpha decay to $^{237}$Np and the subsequent nuclear de-excitation. (Adapted from [3])

$$d_{\text{eff}} = \frac{p}{76\,\text{cm Hg}} \cdot \frac{288\,d}{T + 273}, \tag{1}$$

where $T$ is the room temperature in °C, $p$ is the chamber pressure in cm Hg, and $d$ is the source–detector separation. The reference temperature of 288 K and pressure of 76 cm Hg correspond to the ISA conditions.

### B. $\alpha - \gamma$ coincidence and Nuclear lifetimes

Radioactive decay is the transformation of an unstable nucleus into a more stable one through the emission of particles or radiation. The daughter nucleus may subsequently de-excite by emitting gamma radiation or through other nuclear processes. In the present experiment, $^{241}$Am undergoes alpha decay to $^{237}$Np. The alpha-particle energy depends on the final nuclear state populated in the daughter nucleus, whose subsequent de-excitation can produce gamma photons of characteristic energies. Detecting both radiations provides information about the decay scheme and successive nuclear transition.

The decay scheme of $^{241}$Am is shown in the corresponding figure. Approximately 84% of its alpha decays populate the excited state of $^{237}$Np, with the corresponding alpha particle having an energy of approximately 5.486 MeV. The excited daughter nucleus can subsequently de-excite through different nuclear transitions, including the emission of a 59.5 keV gamma photon.

Not all nuclear de-excitations produce gamma photons. The nucleus may instead transfer its excess energy to an atomic electron, ejecting it from the atom. This process, known as *internal conversion*, competes with gamma-ray emission. Consequently, the total probabilities of all possible gamma emissions do not necessarily add up to unity in the decay scheme. While excited nuclear states generally have lifetimes of the order of picoseconds, states with much longer lifetimes, typically nanoseconds or more, are often called metastable states. The 59.5 keV excited state of $^{237}$Np has a half-life of approximately 67.2 ns.

The principle of alpha–gamma coincidence is to identify alpha and gamma events associated with the same nuclear decay sequence by examining their relative detection times. The alpha detector registers the charged particle, while the gamma detector registers the photon emitted during the subsequent nuclear de-excitation. The electrical signals from both detectors are processed to determine their occurrence times. If the time difference falls within a specified coincidence window, the events are registered as a coincidence. However, unrelated alpha and gamma events may also occur within this window, producing *accidental coincidences*. A correlation heat map is used to distinguish correlated events from uncorrelated ones [3].

For a radioactive nuclear state that decays exponentially with mean lifetime $\tau$, the number of surviving excited nuclei satisfies

$$N(t) = N(0)e^{-t/\tau}. \tag{2}$$

The half-life $t_{1/2}$, defined as the time required for the number of undecayed nuclei to decrease to half its initial value, is related to the mean lifetime by

$$t_{1/2} = \tau \times ln(2). \tag{3}$$

Thus, fitting the time-difference distribution of true coincidence events to an exponential decay function allows the mean lifetime and half-life of the excited state to be determined.

## III. EXPERIMENTAL SET-UP:

#### 1. *Range of alpha particles*

A $^{241}$Am source was used to emit alpha particles, which were detected using a silicon surface-barrier semiconductor detector. When an alpha particle enters the detector, it produces electron–hole pairs in the semiconductor. The applied electric field separates these charge carriers, generating an electrical pulse whose amplitude is proportional to the energy deposited by the particle.

The radioactive source and detector were placed inside a vacuum tube. The pressure inside the tube was varied using a vacuum motor and monitored with a vacuum gauge. The small electrical signals from the detector were processed using a PHYWE preamplifier and a PHYWE multichannel analyser (MCA) operated as an amplifier. The preamplifier converts the small detector signal into a suitable voltage signal for further processing, while the

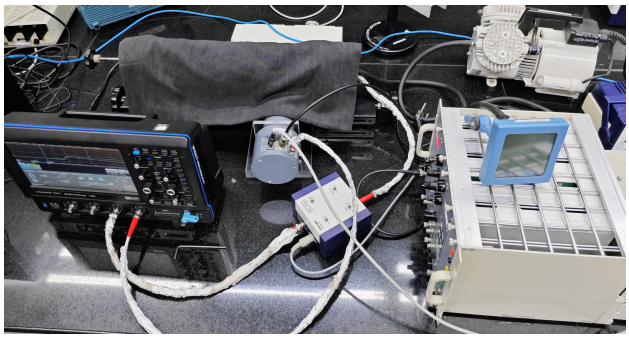


FIG. 3. Experimental setup for $\alpha - \gamma$ coincidence.

amplifier increases the signal amplitude and shapes the pulse for subsequent pulse-height analysis.

The amplifier gain of the pulse-height analyser was adjusted digitally using the PHYWE Measure software. The PHYWE MCA was also used to provide the required power supply to the preamplifier. The Cspark MCA open-source software was then used to acquire and analyse the alpha-particle pulse-height spectrum.

#### 2. *Alpha–gamma coincidence*

For the alpha–gamma coincidence experiment, the same silicon surface-barrier semiconductor detector was used to detect alpha particles, while a $2'' \times 2''$ flat-type NaI scintillation detector (SD152F) was used to detect gamma photons. The gamma detector was operated using a high-voltage power supply at 600 V, while the alpha detector was powered using the PHYWE MCA.

The electrical signal from the alpha detector was first passed through a preamplifier and then connected to one channel of a Teledyne LeCroy WaveSurfer oscilloscope. The gamma-detector signal was connected directly to the other channel. The oscilloscope's Qualified-trigger option was used to capture events. Event A being $\alpha$ signal and event B being $\gamma$ signal. This settings triggers only when $\alpha$ signal is followed by a $\gamma$ signal within the coincidence window. Amplifiers were not used because they introduce timing jitter and affect the measured time difference between the $\alpha$ and $\gamma$ signals.

During measurements involving the $\alpha$ detector, the detector chamber was covered with black chart sheets and black clothing to prevent external light from reaching the detector.

## IV. METHODOLOGY

### A. $\alpha$ particle Range measurement

The range of alpha particles was determined at STP by measuring their residual energy after passage through air. Instead of varying the source–detector separation, the distance between the source and detector was fixed at 3 cm, and the pressure inside the vacuum tube was varied to change the effective distance travelled through air.

Initially, the setup was tested using a LEYBOLD preamplifier together with a PHYWE pulse-height analyser operated as an amplifier. However, neither configuration functioned properly. Therefore, the PHYWE MCA was used as an amplifier for the measurements.

The spectrum-analysis software was calibrated at a pressure below 1 mbar. Data were then acquired for pressure values ranging from below 1 mbar to 510 mbar, with a total of 10 data points. For each pressure, the alpha-particle pulse-height spectrum was recorded, and the corresponding residual-energy peak was identified. The effective distance in standard air was calculated using the measured pressure at room temperature, and fixed source–detector separation.

### B. Attempts with FPGA Data acquisition

A field-programmable gate array (FPGA) is a reconfigurable digital hardware device that can be programmed to perform specific operations, including signal processing, event detection, and precise time measurement. In coincidence experiments, an FPGA can process detector signals and record the time difference between $\alpha$ and $\gamma$ events.

Initially, the coincidence measurements were attempted using the FPGA-based acquisition system. However, several communication issues were encountered. Communication between the Red Pitaya and the computer could not initially be established because the network settings of the two Ethernet interfaces were incompatible. The IP address of the Red Pitaya connection was therefore changed so that it belonged to the same subnet as the computer's network. This enabled a direct Ethernet connection, which was subsequently tested using the Red Pitaya's built-in oscilloscope.

The data-acquisition code available in the Github repository [4] was used. Xilinx Vivado 2015.4 was installed to work with the FPGA source files required for the acquisition setup. However, since the required Vivado licence was unavailable, the precompiled FPGA bitstream provided in the repository was used instead of compiling the source files.

As the Red Pitaya runs Alpine Linux, the `agc.cpp` source file was compiled on Ubuntu to generate an executable compatible with the acquisition setup. The executable and FPGA bitstream were then transferred to the Red Pitaya.

The average $\alpha$-signal amplitude observed on the oscilloscope was approximately 140 mV. The Red Pitaya operates over a range of 0–20 V in high-voltage mode. Since the acquisition system uses 8191 channels, the alpha pulse-height spectrum would occupy only a small number of channels, resulting in poor pulse-height reso-

lution. To increase the pulse amplitude, the alpha preamplifier was switched from the $\alpha$ setting to the $\beta$ setting, providing 10 times more gain.

The Red Pitaya SD-card filesystem was mounted as read-only. Consequently, files copied to the temporary root filesystem were lost whenever the Red Pitaya was disconnected. The device also repeatedly disconnected during attempts to start data acquisition. Considerable time was spent debugging the acquisition software and checking its different stages, but the problem could not be resolved. Owing to the limited time available, we abandon the use of FPGA, and use the Teledyne LeCroy WaveSurfer oscilloscope for the coincidence measurements.

### C. Oscilloscope based data acquisition

For the oscilloscope-based coincidence measurements, the source–$\alpha$-detector separation was fixed at 0.5 cm, with the source and detector aligned. This configuration was chosen to maximise the geometric efficiency without saturating the $\alpha$-detector. The gamma detector was positioned such that the distance between the source and the centre of the NaI crystal was approximately 6–7 cm, as this arrangement provided the maximum gamma count rate.

Throughout the experiment, the pressure inside the alpha-detector chamber was monitored every 3–4 hours and maintained between 1 and 10 mbar. This pressure range corresponded to residual alpha energies between 5433.01 keV and 5424.11 keV.

The oscilloscope's Qualified-trigger configuration was used to capture coincidence events. Event A was assigned to the $\alpha$ signal, while event B was assigned to the $\gamma$ signal. The trigger condition was configured to register an event when an $\alpha$ signal was followed by a $\gamma$ signal within a coincidence window of 600 ns.

Measurements were performed using 3 acquisition configurations. In the first configuration, the total number of samples was set to 50 kS, the sampling rate was 500 MS/s, and the total acquisition time window was 100 $\mu$s. To improve the timing resolution we changed it to the second configuration i.e. 200 kS at a sampling rate of 2 GS/s, with 100 $\mu$s acquisition window. However, the resulting files were too large to handle conveniently. Therefore, the third configuration used 40 kS at the same sampling rate of 2 GS/s, with the acquisition window reduced to 20 $\mu$s.

For all 3 configurations, the $\alpha$ trigger threshold was maintained at 0.6 V, and the $\gamma$ trigger threshold was maintained at 2.3 V. A continuous measurement was carried out for **96** hours, resulting in a total of **4963** recorded coincidence events. Following data acquisition, separate data was acquired for $\alpha$ and $\gamma$ signals for calibration of pulse height with their known energies.

## V. DATA ANALYSIS

### A. Range of $\alpha-$particles

We plot the residual energy of alpha particle against the effective distance in STP caluclated using the Eq. (1). A linear fit was performed to calculate the extrapolated range using the x-intercept as shwon in Fig. 4. The range of alpha particles was calculated to be:

$$R_\alpha = (2.61 \pm 0.01)\ \text{cm} \tag{4}$$

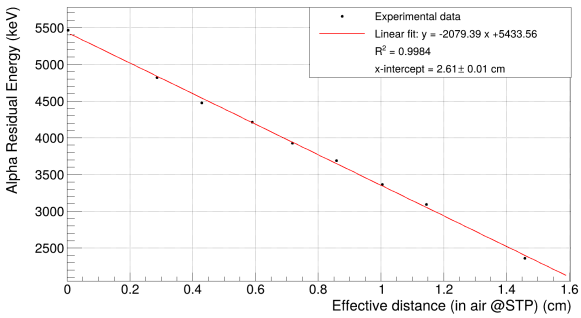



FIG. 4. Measured residual energy of $\alpha$-particles as a function of effective distance in air at STP. The x-intercept of the linear fit is the extrapolated range.

### B. Pulse Height Calibration

For the $\alpha-\gamma$ coincidence experiment, we calibrate the pulse height of $\alpha$ and $\gamma$ signals with the known energies. The pulse height spectrum of $\alpha$ signal was fitted with a Gaussian with the mean pulse height was calibrated with $E_\alpha = 5.486$ MeV as shown in Fig. 5. Also, the photopeak in $\gamma$ pulse height spectrum was fitted with the Gaussian and calibrated with the gamma energy $E_\gamma = 59.5$ keV as shown in Fig. 6. From this analysis, the following energy calibration relations were obtained,

$$E_\alpha = 7.63\ V_\alpha\ \text{MeV} \tag{5}$$
$$E_\gamma = 20.05\ V_\gamma\ \text{keV} \tag{6}$$

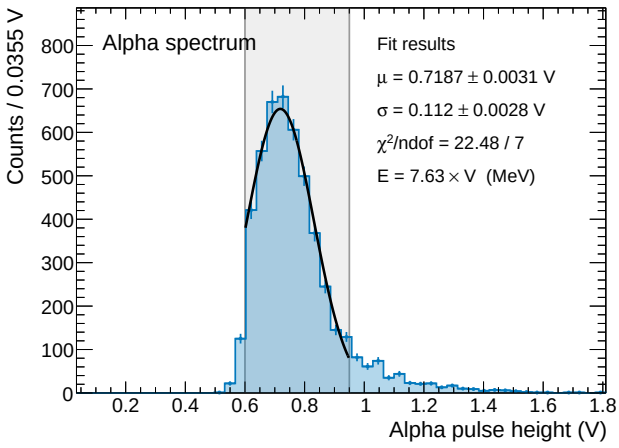



FIG. 5. Alpha-particle pulse-height spectrum with a Gaussian fit to the peak. The fitted mean is used to calibrate the pulse height using the known alpha-particle energy of 5.486 MeV.

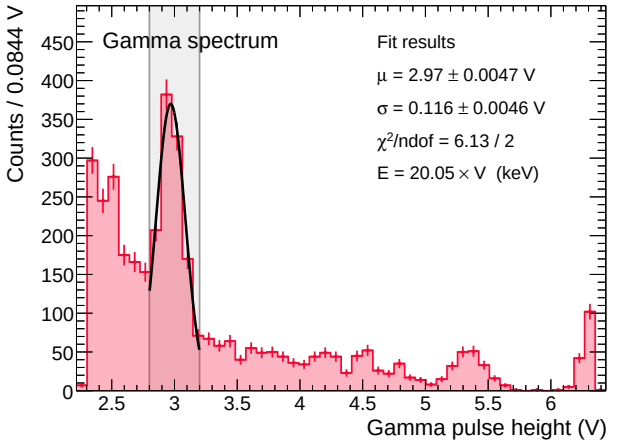



FIG. 6. Gamma-ray pulse-height spectrum with a Gaussian fit to the 59.5 keV photopeak.

### C. Baseline RMS Filter

The $\alpha - \gamma$ coincidence events stored in `.csv` files were too large to analyse, and our parameters of interest lied in a small range, a region of interest is defined near the trigger time. A window from $-4\mu s$ to $+4\mu s$ around the trigger time is chosen as a region of interest. We read only data points only in this region and extract parameters like baseline, $\alpha$ and $\gamma$ pulse height, peak time, trigger time and constant-fraction time discussed in the next section.

Some of the files were found to have a few overlapping events i.e. the triggered event signal was overlapping with a previous signal. This hinders our ability to estimate its peak accurate since its pulse height also has a contribution from the previous uncorrelated event. We use a baseline RMS cutoff technique to filter out these noisy events. The baseline mean and RMS values were calculated in the range $-4\mu s$ to $-2\mu s$. The RMS cut-off to filter out signals was decided using the RMS distribution attached in supplementary material. The events with a baseline RMS less than the cut-off were considered clean signals and used for further analysis.

### D. Energy Correlation Heatmap

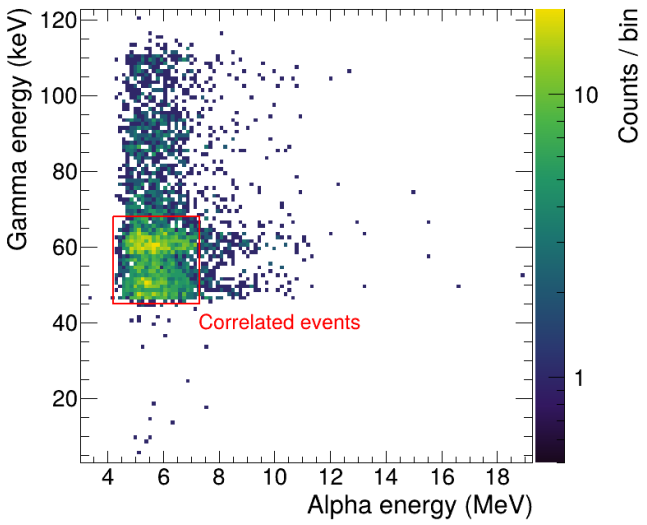



FIG. 7. Log-scale correlation heatmap of calibrated alpha-particle energy and gamma-ray energy. The correlated-event region (red) is selected using the expected energies from the $^{241}$Am decay scheme.

We convert the pulse heights of signals to corresponding energies using the calibration Eq. (6). To isolate events from the same decay scheme, we plot the correlation heatmap between alpha signal energy ($E_\alpha$ and gamma signal energy ($E_\gamma$) as shown in Fig. 7. The counts are denoted by color (log-scale). The linear scale plot is attached in supplementary material. We choose the region of correlated events according to the known correlated $\alpha$ and $\gamma$ energies from the Am-241 decay scheme. These correlated events are used for further analysis.

### E. Time-walk correction

The time difference between the signals was quantified in the following 3 different ways:

$$\Delta t_{Peak} = t_{Peak}(\gamma) - t_{Peak}(\alpha) \quad (7)$$

$$\Delta t_{Trigger} = t_{Trigger}(\gamma) - t_{Trigger}(\alpha) \quad (8)$$

$$\Delta t_{CF} = t_{CF}(\gamma) - t_{CF}(\alpha) \quad (9)$$

$$(10)$$

where $t_{Peak}$ denotes the time of the peak, $t_{Trigger}$ denotes time of trigger of event according to the threshold set during measurements. The $t_{CF}$ denotes Constant fraction time i.e. the time taken for the signal to achieve a fraction of its pulse height. The constant fraction was chosen to be 20%. This way of quantifying arrival time of the signal proves to be useful later in the report. We observed a plateau region in $\Delta t$ distributions which is generally caused due to time walk systematic shifts.

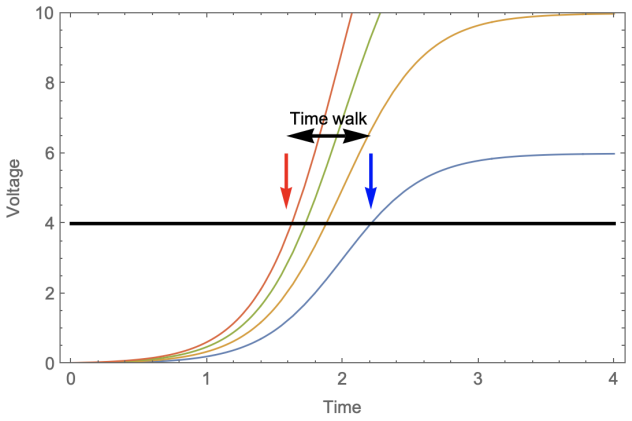



FIG. 8. Schematic of time walk, showing the dependence of the measured signal arrival time on pulse amplitude.

Timing walk is a systematic dependence of the measured time on pulse amplitude. The detector and electronics respond slowly, so a bigger pulse reaches the same timing point at a different moment, even when the physical emission time is identical as shown schematically in Fig. 8. This systematic shift smears the exponential distribution of $\Delta t$ and gives a plateau.

We plot the heatmaps of $\Delta t$ vs pulse height of $\alpha$ and $\gamma$ signal as shown in the left plots in Fig. 11, 12, 13. The red points in the plots denote the mean $\Delta t$. We observe the expected pulse height dependence of $\Delta t_{mean}$. We model this variation using,

$$\Delta t_{mean} = c + f(V_\alpha) + h(V_\gamma) \tag{11}$$

where $c$ = constant, $f$ and $h$ are polynomial of degree 2. This systematic pulse height dependence is then subtracted from the calculated $\Delta t$ and the corrected distribution is obtained as shown in the right plots in Fig. 11, 12, 13.

### F. Half life of $^{237}$Np excited state

We plot the distribution of $\Delta t$ for the filtered signals. The $\Delta t_{Peak}$ plot even after correction did not exhibit an exponential decay region and is attached in the supplementary material. The $\Delta t_{Trigger}$ and $\Delta t_{CF}$ distributions exhibited an exponential decay region highlighted in the

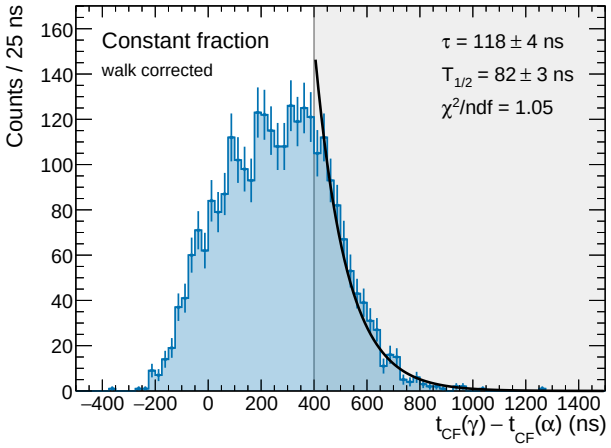



FIG. 9. Distribution of $\Delta t_{CF}$, after time-walk correction. The shaded region is fitted with an exponential decay function to determine the mean lifetime of the excited $^{237}$Np state.

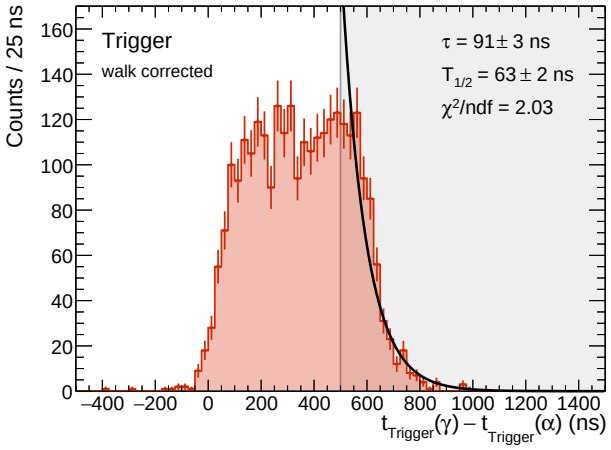



FIG. 10. Distribution of $\Delta t_{Trigger}$, after time-walk correction. The shaded region is fitted exponentially to estimate the mean lifetime and half-life of the excited $^{237}$Np state.

Fig. 10 and Fig. 9 respectively. The shaded regions of these plots were exponentially fitted to obtain mean lifetime and half-life of $^{237}$Np excited state using Eq. (3).

Mean lifetime:

$\tau = (91 \pm 3)$ ns (using Trigger time)

$\tau = (118 \pm 4)$ ns (using CF time)

Half-life:

$T_{1/2} = (63 \pm 2)$ ns (using Trigger time)

$T_{1/2} = (82 \pm 3)$ ns (using CF time)

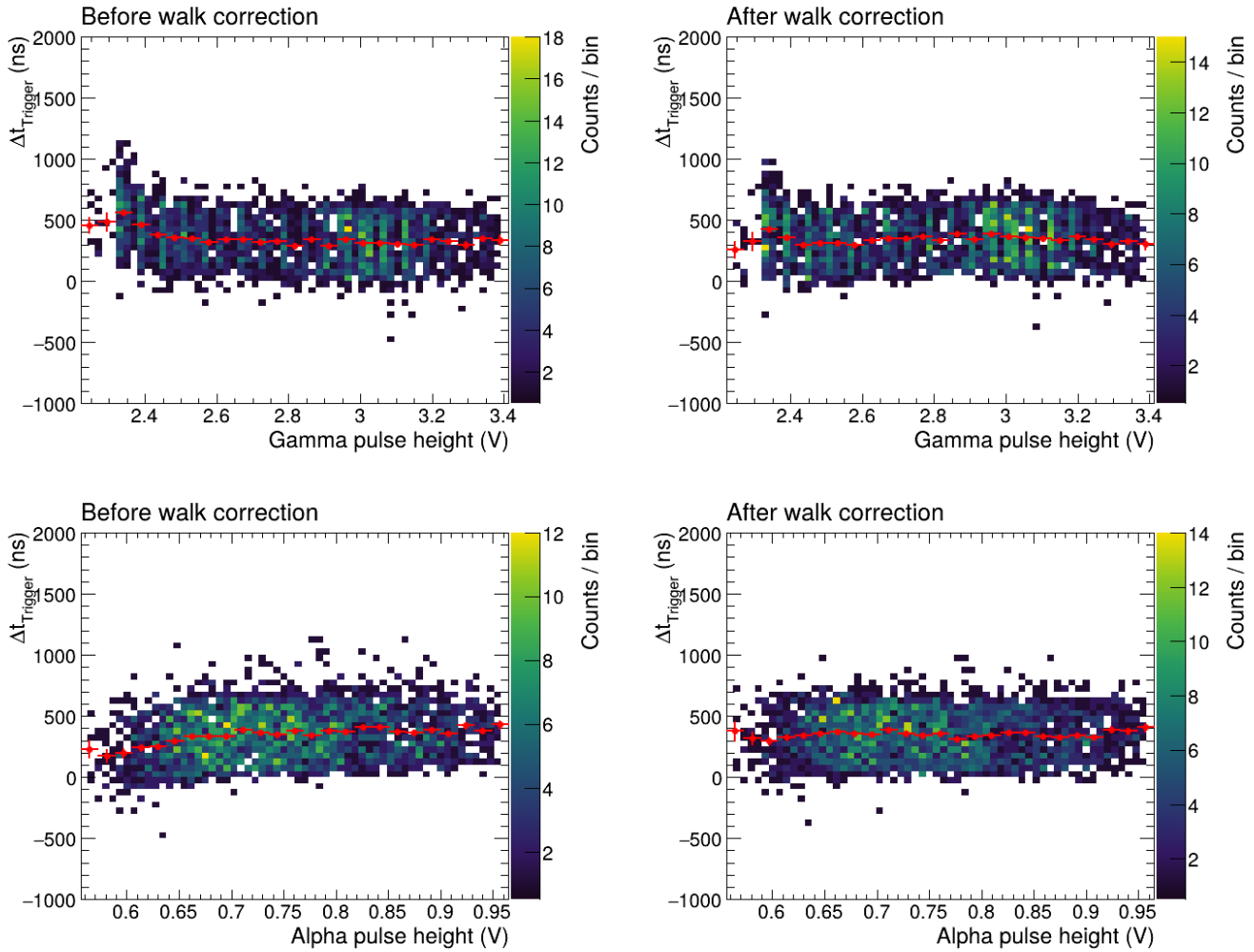



FIG. 11. Trigger-time walk correction. The left panels show the dependence of the mean trigger time difference on $\alpha$ pulse heights (bottom) and $\gamma$ pulse heights (top). The right panel shows the time-difference distribution after applying the correction.

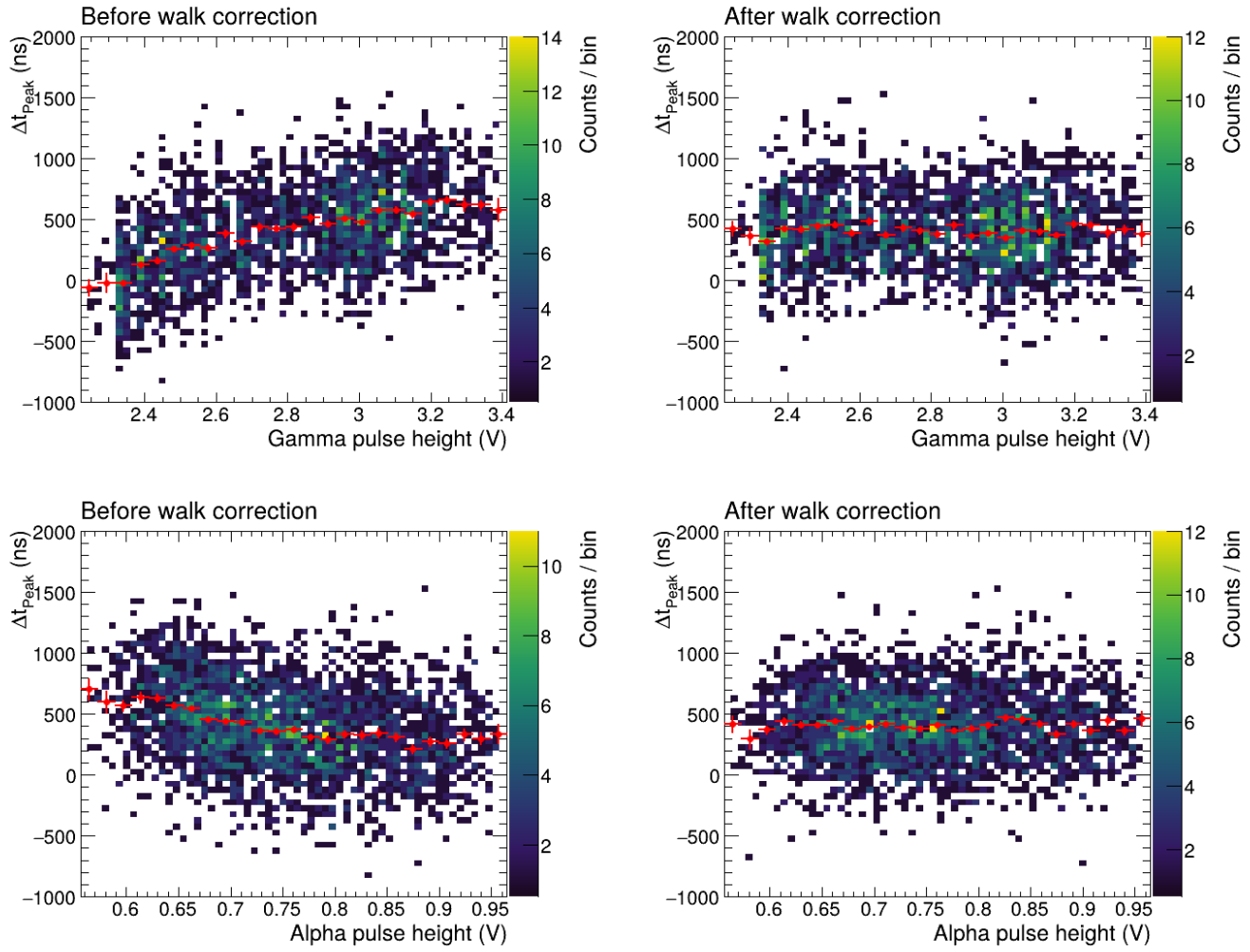



FIG. 12. Peak-time walk correction. The left panels show the dependence of the mean peak time difference on $\alpha$ pulse heights (bottom) and $\gamma$ pulse heights (top). The right panel shows the time-difference distribution after applying the correction.

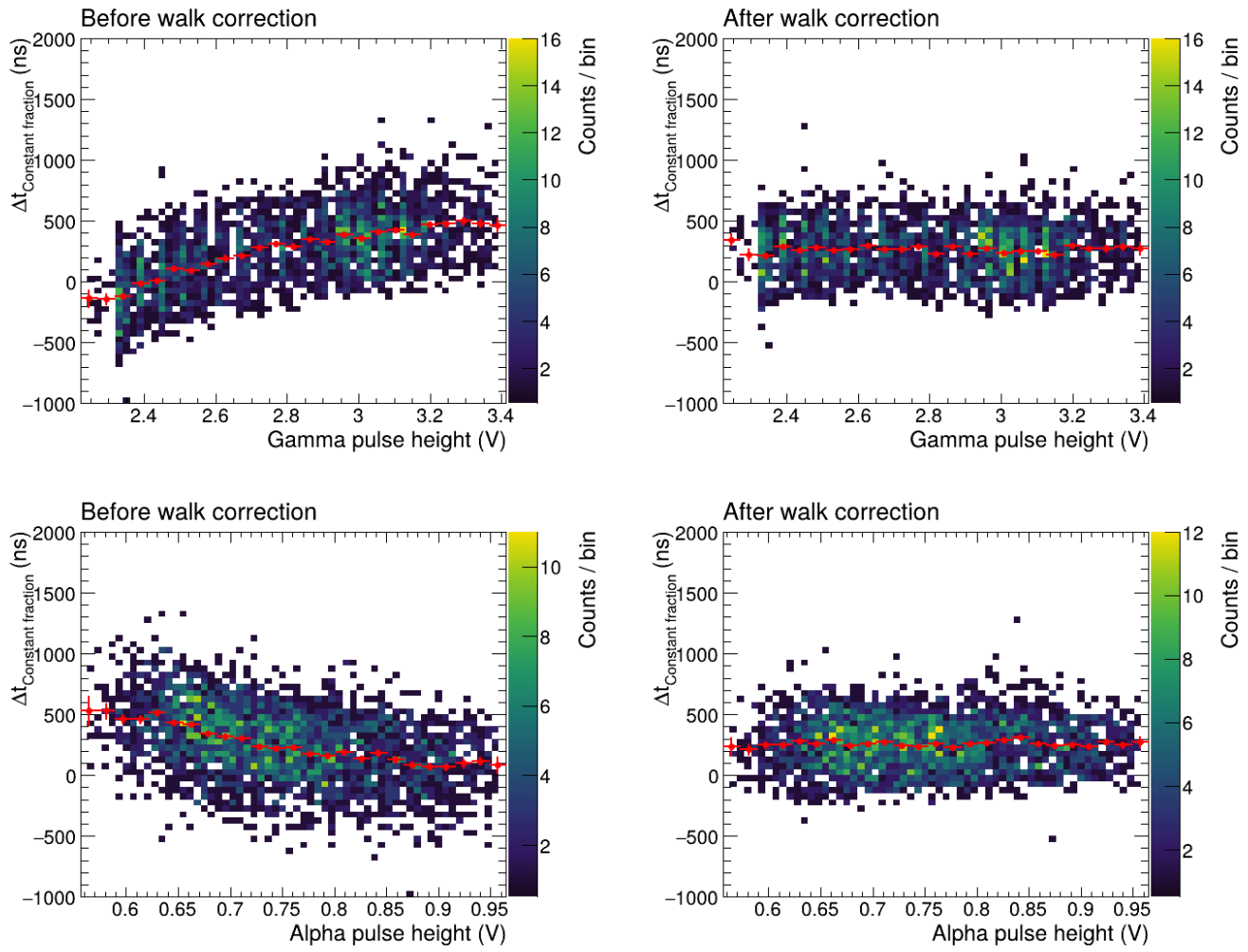



FIG. 13. Constant fraction-time walk correction. The left panels show the dependence of the mean constant fraction time difference on $\alpha$ pulse heights (bottom) and $\gamma$ pulse heights (top). The right panel shows the time-difference distribution after applying the correction.

## VI. RESULTS AND DISCUSSION

The residual alpha energy decreased with increasing effective distance in air. A linear fit to the data (Fig. 4) gave an extrapolated range of

$$R_\alpha = (2.61 \pm 0.01)\,\text{cm}$$

The alpha and gamma pulse-height spectra were calibrated using the known energies of 5.486 MeV and 59.5 keV, respectively (Figs. 5 and 6). The resulting calibration relations were

$$E_\alpha = 7.63\ V_\alpha\ \text{MeV}$$
$$E_\gamma = 20.05\ V_\gamma\ \text{keV}$$

The energy-correlation heatmap (Fig. 7) was used to select events consistent with the expected decay scheme.

A total of 4963 coincidence events were recorded over continuous measurement of 96 hours. After correcting for pulse-height-dependent time walk (Figs. 11, 12, 13), the trigger-time and constant-fraction distributions (Figs. 10, 9) yielded the following half-lives and corresponding mean lifetimes:

$$T_{1/2}^{\text{trigger}} = (63 \pm 2)\,\text{ns}, \qquad \tau^{\text{trigger}} = (91 \pm 3)\,\text{ns},$$
$$T_{1/2}^{\text{CF}} = (82 \pm 3)\,\text{ns}, \qquad \tau^{\text{CF}} = (118 \pm 4)\,\text{ns}.$$

The trigger-time result is closer to the reference half-life of approximately 67.2 ns. The difference between the two estimates indicates residual timing-related systematic effects and dependence on the timing method.

The trigger-time result is closer to the reference value of approximately 67.2 ns.

## VII. CONCLUSION

The alpha-particle range in air was estimated as $(2.61 \pm 0.01)$ cm at STP. Baseline RMS cut-offs were used to filter noisy events. $E_\alpha - E\gamma$ correlations enabled the selection of correlated decay events. A timewalk correction was applied to these events. The exponential region of timewalk corrected distributions was fitted to obtain mean lifetimes. The trigger-time method yielded a half-life closer to the reference value than the constant-fraction method. Improved timing corrections and systematic uncertainty estimates are needed for more precise lifetime measurements.

## VIII. ACKNOWLEDGEMENT

We thank Purvesh Garge, a PhD student in the Quantum Optics Lab, for help in identifying the IP-address mismatch as the cause of the FPGA communication problem. Also we acknowledge Claude for writing the C++ code used in the data analysis.

## IX. CODE AND DATA AVAILABILITY

Analysis code, plotting scripts, experimental data, and supplementary material are available in the GitHub repository.

[1] *Range of Alpha Particles*, October 2014. Laboratory Instruction Manual.

[2] William R. Leo. *Techniques for Nuclear and Particle Physics Experiments: A How-to Approach*. Springer, Berlin, Heidelberg, 2 edition, 1994.

[3] Mario Vretenar, Nataša Erceg, and Marin Karuza. Energy-resolved coincidence counting using an fpga for nuclear lifetime experiments. *American Journal of Physics*, 87(12):997–1002, 2019.

[4] Mario Vretenar, Nataša Erceg, and Marin Karuza. Agc: Alpha-gamma coincidence data acquisition system for Red Pitaya, 2019.